

ABDUL SAMHOON

Skills

- Data Analysis (Cleaning, EDA, Feature Engineering)
- Python (Pandas, NumPy, Scikit-learn, Matplotlib)
- Microsoft Power BI (Interactive Dashboards)
- Excel (VLookup, Conditional Formatting, Pivot Tables)
- MySQL
- Tools (Git & GitHub, VS Code, Google Colab)
- Soft Skills (Presentation, English Communication)

Projects

EARLY WARNING SYSTEM FOR URBAN POLLUTION SPIKES — *PERSONAL PROJECT*

January 2026

- Developed a ML-based early warning system to forecast maximum NO2 concentration for 24-hour windows.
- Engineered time-series features including lagged pollutant values and spike indicators while preventing data leakage.
- Trained and evaluated a Random Forest regressor, achieving ~32 MAE and enabling proactive urban planning decisions.

SMART PATHWAYS FOR PERSONALIZED LEARNING — *ACADEMIC TEAM PROJECT*

November 2025

- Developed an AI platform for structured learning paths and curated career resources.
- Integrated Groq API for automated quiz generation and real-time learner evaluation.
- Built mock interview and networking features to enhance student job-readiness.

HOSPITAL SURGE PREDICTION — *HACKATHON TEAM PROJECT*

August 2025

- Built a Random Forest model to forecast patient inflow surges using real hospital data.
- Executed end-to-end data cleaning and EDA to identify seasonal demand patterns.
- Visualized predictive insights to optimize hospital staffing and capacity management.

RESTAURANT RESERVATION SYSTEM — *HACKATHON TEAM PROJECT*

April 2026

- Built a full-stack restaurant reservation system with table layout, real-time availability, and time-based booking.
- Implemented pricing based on seats and duration with integrated payment simulation.
- Developed admin dashboard for table control, reservations tracking, and receipt generation with token system.

Awards & Certifications

- 2nd Prize: AI Sparkathon 2025 winner at Bearys Institute of Technology.
- 9th Place: National Level Hackathon participant and top-tier finalist.
- DeepLearning.AI: Supervised Machine Learning (Regression & Classification) certification.
- Hackathons: Participated in AXIOM State-level Hackathon, Google Agent AI Day, and InUnity Innovation for Community.

Education

BEARYS INSTITUTE OF TECHNOLOGY, Mangalore — Bachelor of Engineering in Artificial Intelligence and Data Science 2023 – 2027

CGPA: 8.6

Other Activities

- Participated in college technical and data science club activities and workshops.
- Attended hackathons, tech talks, and industry webinars to enhance practical skills.
- Collaborated on team projects and peer learning initiatives in data analysis and machine learning.